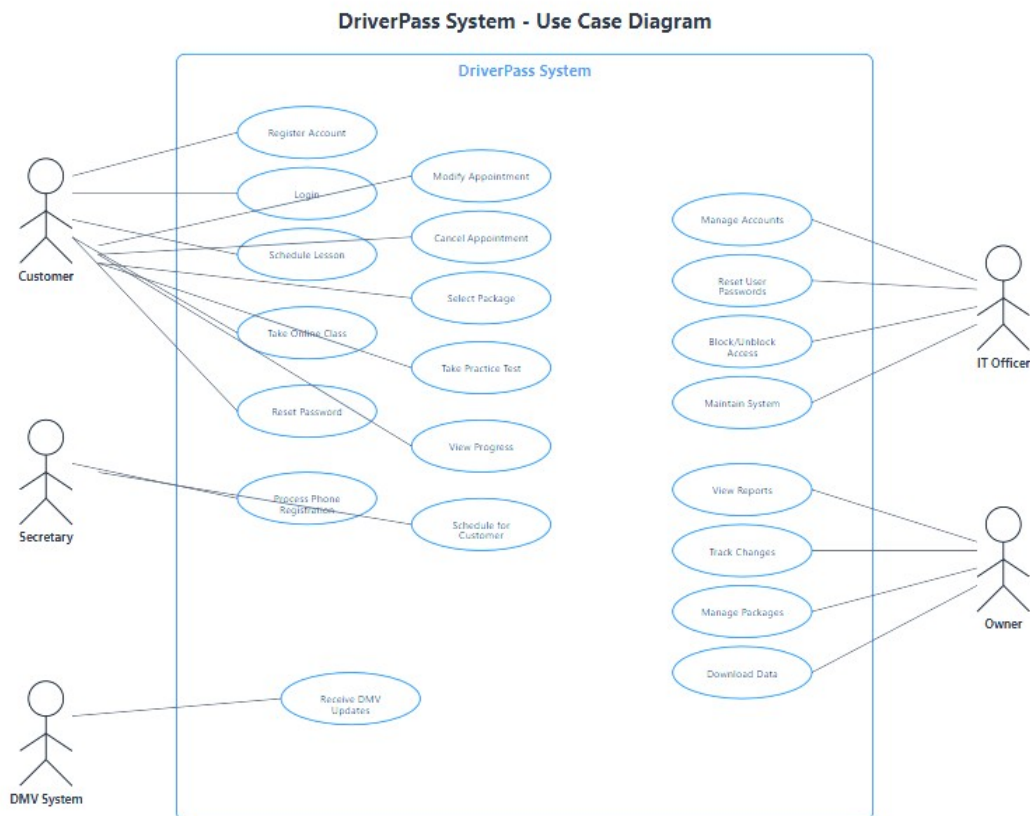


CS 255 System Design Document Template

This template lays out all the different sections that you need to complete for Project Two. Each section has guidance to prompt your thinking. You will need to continually reference the interview transcript as you work to make sure that you are addressing your client's needs. There is no required length for the final document. Instead the goal is to complete each section based on what your client's needs are. Remove this note when you are finished, and replace all bracketed text with the relevant information.

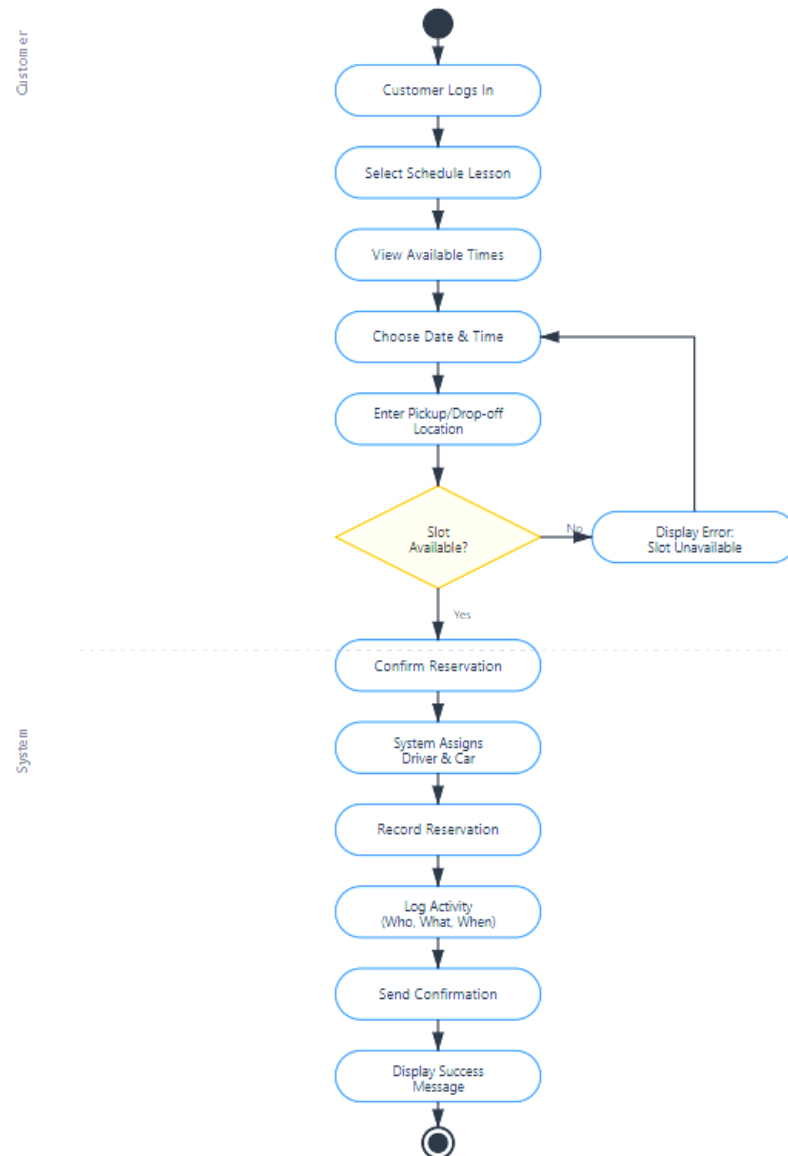
UML Diagrams

UML Use Case Diagram



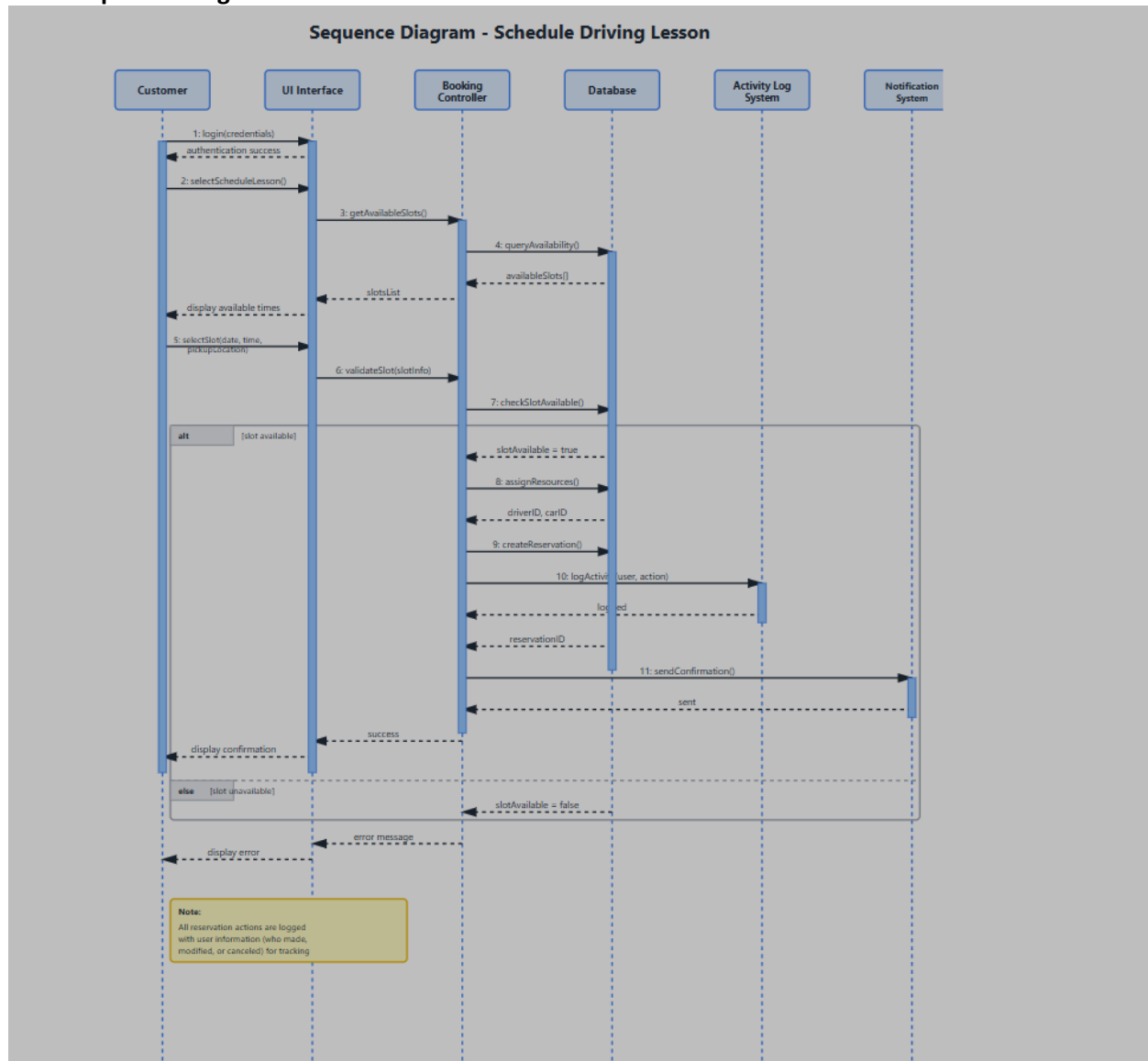
UML Activity Diagrams

Activity Diagram - Schedule Driving Lesson

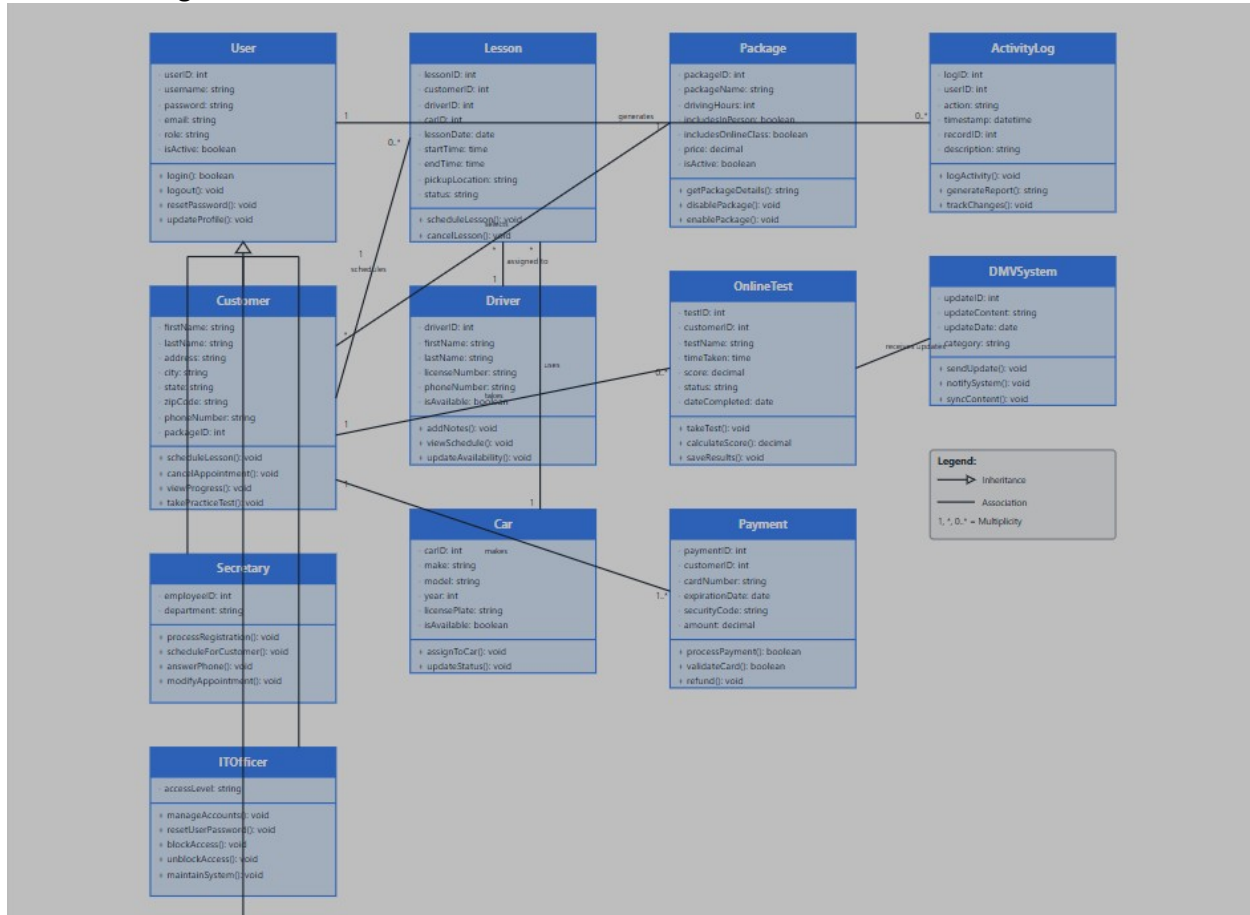


[

UML Sequence Diagram



UML Class Diagram



Technical Requirements

Hardware Requirements

Server Infrastructure

- **Cloud-based servers** (AWS, Azure, or Google Cloud Platform recommended)
 - Minimum 4-core CPU processors
 - 16 GB RAM minimum (32 GB recommended for production)
 - 500 GB SSD storage minimum (scalable based on user growth)
 - Load balancers for traffic distribution
 - Redundant servers for high availability and disaster recovery
 - Auto-scaling capabilities to handle traffic spikes

Client Devices

- **Desktop computers:** Windows 10/11, macOS 10.14+, or Linux

- **Mobile devices:** iOS 13.0+ and Android 8.0+
- **Minimum client specifications:**
 - 4 GB RAM
 - 100 MB available storage
 - Internet connection: minimum 5 Mbps

Database Server

- **Dedicated database servers** with RAID 5 or RAID 6 configuration
- Minimum 250 GB storage for initial deployment
- Automatic backup systems (daily backups minimum)
- Database replication for redundancy

Software Requirements

Backend Technology Stack

- **Programming Language:** Java, Python, or C# (.NET)
- **Web Framework:** Spring Boot (Java), Django/Flask (Python), or ASP.NET Core (C#)
- **Application Server:** Apache Tomcat, Nginx, or IIS
- **API Protocol:** RESTful API with JSON/XML support

Database Management System

- **Primary Database:** PostgreSQL or MySQL
 - Support for ACID transactions
 - Full-text search capabilities
 - JSON data type support
- **Caching Layer:** Redis for session management and performance optimization
- **Backup System:** Automated daily backups with off-site replication

Frontend Technology Stack

- **Web Application:** HTML5, CSS3, JavaScript (ES6+)
- **Frontend Framework:** React.js, Angular, or Vue.js
- **Mobile Applications:**
 - iOS: Swift or React Native
 - Android: Kotlin or React Native
- **Responsive Design:** Mobile-first approach for all interfaces

Security Infrastructure

- **SSL/TLS:** HTTPS/TLS 1.2 or higher for all communications
- **Authentication:** OAuth 2.0 or JWT (JSON Web Tokens)
- **Authorization:** Role-Based Access Control (RBAC)
- **Encryption:** AES-256 encryption for sensitive data at rest
- **Password Management:** bcrypt or Argon2 for password hashing

Third-Party Integrations

- **Payment Gateway:** Stripe, PayPal, or Square API
- **Email Service:** SendGrid, AWS SES, or Mailgun
- **SMS Notifications:** Twilio
- **DMV Integration:** Custom API endpoints for receiving updates

Software Tools and Development Environment

Development Tools

- **Version Control:** Git with GitHub, GitLab, or Bitbucket
- **IDE/Code Editors:** Visual Studio Code, IntelliJ IDEA, or JetBrains Suite
- **Build Tools:** Maven, Gradle, or npm/webpack
- **Continuous Integration:** Jenkins, GitLab CI, or GitHub Actions

Testing Tools

- **Unit Testing:** JUnit (Java), pytest (Python), or xUnit (.NET)
- **Integration Testing:** Postman, REST Assured, or SoapUI
- **Automated Testing:** Selenium, Cypress, or TestNG
- **Performance Testing:** Apache JMeter or LoadRunner
- **Test Coverage:** SonarQube for code quality analysis

Deployment and DevOps

- **Containerization:** Docker for application containerization
- **Orchestration:** Kubernetes for container management
- **CI/CD Pipeline:** Automated build, test, and deployment processes
- **Monitoring:** Prometheus, Grafana, or ELK Stack (Elasticsearch, Logstash, Kibana)
- **Logging:** Centralized logging system (Log4j, Serilog, or Winston)

Documentation Tools

- **API Documentation:** Swagger/OpenAPI
- **Project Management:** Jira, Asana, or Monday.com
- **Collaboration:** Confluence, Notion, or SharePoint

Infrastructure Requirements

Networking

- **Internet Connectivity:** Redundant internet connections (minimum 1 Gbps)
- **VPN:** Virtual Private Network for secure remote access
- **Firewall:** Enterprise-grade firewalls with DDoS protection
- **Content Delivery Network (CDN):** CloudFlare or AWS CloudFront for static content delivery
- **DNS:** Managed DNS service with high availability

Cloud Services Configuration

- **Region Selection:** Multi-region deployment for disaster recovery
- **Availability Zones:** Deployment across multiple availability zones
- **Auto-scaling:** Auto-scaling groups configured based on traffic metrics
- **Content Delivery:** Global CDN configuration for fast content delivery
- **Object Storage:** AWS S3, Azure Blob Storage, or Google Cloud Storage for file storage

Backup and Disaster Recovery

- **Backup Frequency:** Daily automated backups with retention policy (minimum 30 days)
- **Recovery Time Objective (RTO):** 4 hours maximum
- **Recovery Point Objective (RPO):** 1 hour maximum
- **Disaster Recovery Site:** Geographically distant backup infrastructure
- **Testing:** Quarterly disaster recovery drills and testing

Compliance and Security Standards

Data Protection

- **GDPR Compliance:** For customers in the European Union
- **CCPA Compliance:** For customers in California
- **PCI DSS Compliance:** Payment Card Industry standards for payment processing
- **Data Encryption:** End-to-end encryption for sensitive data transmission

Access Control

- **Role-Based Access Control (RBAC):** Four primary roles (Customer, Secretary, IT Officer, Owner)
- **Audit Trails:** Complete logging of all user actions and data modifications
- **Session Management:** Automatic session timeout after 30 minutes of inactivity
- **Password Policy:** Minimum 12 characters, complexity requirements, 90-day expiration

Network Security

- **Web Application Firewall (WAF):** Protection against common web vulnerabilities
- **Intrusion Detection:** Real-time threat detection and response
- **Penetration Testing:** Annual security audits and penetration testing
- **SSL Certificates:** Valid SSL/TLS certificates with annual renewal

Performance Requirements

Response Time

- **Page Load Time:** Maximum 3 seconds for desktop, 5 seconds for mobile
- **API Response Time:** Maximum 500ms for 95th percentile requests
- **Database Query Time:** Maximum 100ms for standard queries

Availability

- **System Uptime:** 99.5% availability (approximately 3.6 hours downtime per month)
- **Maintenance Windows:** Scheduled weekly maintenance during off-peak hours
- **Monitoring:** 24/7 system monitoring with automated alerts

Scalability

- **Concurrent Users:** Support minimum 10,000 concurrent users
- **Database Growth:** Support database growth to 1TB within 5 years
- **Transaction Throughput:** Minimum 1,000 transactions per second capacity

Maintenance and Support

System Monitoring

- **Real-time Monitoring:** 24/7 monitoring of system performance and availability
- **Alerting:** Automated alerts for critical issues
- **Reporting:** Weekly performance reports to management

Regular Maintenance

- **Security Patches:** Monthly security updates and patches
- **Software Updates:** Quarterly framework and library updates
- **Database Maintenance:** Weekly database optimization and maintenance
- **Server Maintenance:** Monthly server health checks and optimization

User Support

- **Help Desk Support:** Email and phone support during business hours
- **Ticketing System:** Issue tracking and resolution system
- **Knowledge Base:** Online documentation and FAQs for common issues

Compliance with DriverPass Business Requirements

These technical requirements directly support DriverPass's business needs:

1. **Cloud-based Accessibility:** Multi-platform support enables access from any device, anywhere
2. **Data Security and Privacy:** Encryption, RBAC, and audit trails protect customer information and credit card data
3. **Availability:** 99.5% uptime ensures customers can schedule lessons and access practice tests
4. **Scalability:** Auto-scaling infrastructure supports growth as DriverPass expands
5. **Integration:** Custom APIs enable seamless DMV system updates
6. **Tracking and Reporting:** Comprehensive logging and audit trails track all user activities
7. **Performance:** Response times ensure smooth user experience for online classes and practice tests
8. **Disaster Recovery:** Multi-region deployment and backup systems protect business continuity